

Foundations of Arithmetic Power Geometry II: Information-Geometric Formulations, Coordinate Stability, and Logarithmic Projective Height Mappings

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Abstract

We present a regularized information-geometric extension of Arithmetic Power Geometry (APG). Moving beyond purely descriptive dictionaries, this work introduces a coordinate-stabilized tensor framework on the continuous Power-Deformation Space $P_{(a,b)}$.

This manuscript establishes five core developments. First, it introduces a stabilized metric formulation, proving that a positive diagonal regularizer λ yields a positive-definite and non-degenerate information tensor field over prescribed exponent domains. Second, it establishes a coordinate stability and extremal entropy theorem, demonstrating analytically that balanced coordinate systems maximize deformation sensitivity at the Euclidean target. Third, it proves a one-sided height–radical entropy bound, linking the APG Global Information Invariant I_n to classical absolute logarithmic Weil projective heights. Fourth, it gives an asymptotic geodesic structure proposition, characterizing the differential equations of motion for moving exponent parameters across the information landscape. Fifth, it states conjectural pathways for asymptotic curvature-growth profiles and Frey-curve/Faltings-height mappings, explored through finite-window computations.

The paper is positioned as a framework paper. It does not claim an independent proof of Fermat’s Last Theorem, the *abc* conjecture, or a proved Frey–Faltings obstruction. The rigorous results are restricted to the stated metric, entropy, and one-sided height-bound assertions; the curvature and Frey/Faltings sections are explicitly conjectural.

Keywords: arithmetic geometry; information geometry; metric tensor; coordinate stability; projective heights; Diophantine approximation.

1 Introduction and Core Philosophy

Classical arithmetic geometry investigates rational or integral points of algebraic varieties defined by polynomial equations over fixed rings, a viewpoint developed throughout modern algebraic and Diophantine geometry [16, 18, 17]. In that paradigm, the exponent structure of the defining equation is fixed, while the coordinate fields vary.

Arithmetic Power Geometry (APG) adopts a complementary viewpoint, continuing the local exponent-deformation program introduced in APG I [1]. It treats the exponent as a continuously varying parameter and studies how algebraic closure relations deform over a parameter landscape. In this framework, base coordinates are fixed positive values, while the exponent operates as a deformation variable.

The motivation arises from the distinction between the Euclidean baseline $n = 2$, where the relation

$$a^2 + b^2 = c^2$$

admits classical geometric closure, and higher-power regimes $n > 2$, where exact integer closure is arithmetically restrictive in the classical Fermat tradition [11]. APG does not replace classical Diophantine geometry. Instead, it provides an information-geometric language for measuring how closure relations deform as the exponent moves away from the Euclidean target.

The present paper extends the local APG framework by introducing a stabilized information tensor, coordinate stability principles, a one-sided height-radical entropy bound, and geodesic structure, drawing on standard information-geometric and Riemannian formulations [23, 3, 2, 10, 19, 21]. The speculative interfaces to curvature growth and Frey/Faltings heights are stated as conjectures rather than theorems.

2 Foundational Space and Local Asymptotics

Let

$$(a, b) \in (\mathbb{R}^+)^2$$

be fixed positive base coordinates. The two-dimensional Power-Deformation Space is the open manifold

$$P_{(a,b)} = \{(c, n) \in \mathbb{R}^+ \times \mathbb{R}^+ : c > 0, n > 0\}.$$

The Euclidean Target is

$$c_2 = \sqrt{a^2 + b^2}.$$

The Local Closure Defect at the Euclidean target is defined by

$$\delta_K(a, b, n) = \left| 1 - \frac{a^n + b^n}{(a^2 + b^2)^{n/2}} \right|. \quad (1)$$

Let

$$K_P = n - 2$$

denote the local deformation parameter. Define the normalized coordinate weights

$$w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}.$$

Their Shannon entropy, measured in nats [28, 8], is

$$H(W) = -(w_a \ln w_a + w_b \ln w_b).$$

The foundational local APG result, established in APG I [1], is the asymptotic expansion

$$\delta_K = \frac{H(W)}{2} |K_P| + O(K_P^2), \quad K_P \rightarrow 0. \quad (2)$$

Remark 1. Equation (2) is a local asymptotic result near $n = 2$. It is not a global obstruction theorem and does not imply Fermat-type non-solvability over the integers.

3 Global Positivity of the Euclidean-Target Defect

Proposition 1 (Global Positivity of the Euclidean-Target Defect). *For every positive pair (a, b) , the Local Closure Defect satisfies*

$$\delta_K(a, b, n) > 0$$

for all real exponents $n > 2$.

Proof. Define

$$f(n) = \frac{a^n + b^n}{(a^2 + b^2)^{n/2}}.$$

The desired assertion is equivalent to proving $f(n) < 1$ for $n > 2$, because $f(2) = 1$.

This follows from monotonicity of ℓ^p -norms, a standard tool in analysis and Diophantine approximation contexts [7, 26]. For a fixed vector (a, b) with positive coordinates, the norm

$$\|(a, b)\|_p = (a^p + b^p)^{1/p}$$

is non-increasing in p . Hence, for $n > 2$,

$$(a^n + b^n)^{1/n} < (a^2 + b^2)^{1/2}$$

whenever both coordinates are positive and $a \neq b$. Raising both sides to the n -th power gives

$$a^n + b^n < (a^2 + b^2)^{n/2}.$$

Therefore $f(n) < 1$, and so

$$\delta_K(a, b, n) = 1 - f(n) > 0.$$

□

Remark 2. *This proposition concerns deformation at the Euclidean Target c_2 . It does not constrain possible closure at arbitrary coordinates $c \neq c_2$, and therefore it is not a proof of any Fermat-type non-solvability theorem.*

4 Convex Potential Information Tensor Formulation

Let local coordinates on $P_{(a,b)}$ be

$$x^1 = c, \quad x^2 = n.$$

Following the potential-based viewpoint used in information geometry and contrast geometry [12, 2], define the potential

$$\Phi(c, n) = -\ln \left(\frac{c^n}{a^n + b^n} \right) = \ln(a^n + b^n) - n \ln c.$$

To obtain a positive-definite information tensor over a prescribed exponent domain, introduce a positive diagonal regularizer $\lambda > 0$. Define

$$g_{ij} = \frac{\partial^2 \Phi}{\partial x^i \partial x^j} + \lambda \delta_{ij}.$$

A direct computation yields

$$g_{cc} = \frac{n}{c^2} + \lambda, \quad g_{cn} = g_{nc} = -\frac{1}{c}, \quad g_{nn} = V_n + \lambda, \tag{3}$$

where

$$V_n = \frac{a^n b^n (\ln(a/b))^2}{(a^n + b^n)^2}.$$

Theorem 1 (Positive Definiteness on a Stabilized Exponent Domain). *Let $n \geq n_0 > 0$. If*

$$\lambda > \frac{1}{n_0},$$

then the tensor $g = (g_{ij})$ defined by (3) is positive definite on the domain $n \geq n_0$. In particular, on the APG deformation domain $n \geq 2$, it suffices to choose

$$\lambda > \frac{1}{2}.$$

Proof. For any nonzero vector $v = (v_1, v_2) \in \mathbb{R}^2$,

$$v^T g v = \left(\frac{n}{c^2} + \lambda \right) v_1^2 - \frac{2}{c} v_1 v_2 + (V_n + \lambda) v_2^2.$$

Completing the square gives

$$v^T g v = \left(\frac{\sqrt{n}}{c} v_1 - \frac{1}{\sqrt{n}} v_2 \right)^2 + \left(V_n + \lambda - \frac{1}{n} \right) v_2^2 + \lambda v_1^2.$$

Since $V_n \geq 0$ and $n \geq n_0$, we have

$$\frac{1}{n} \leq \frac{1}{n_0}.$$

Thus $\lambda > 1/n_0$ implies

$$V_n + \lambda - \frac{1}{n} > 0.$$

The first term is nonnegative, and the remaining terms make the quadratic form strictly positive for every nonzero v . Hence g is positive definite. $\square$

5 Asymptotic Curvature Growth Profile

Conjecture 1 (Asymptotic Curvature Growth Profile). *For the regularized APG information tensor g , the scalar curvature $R_{\text{APG}}(c, n)$ satisfies the asymptotic growth profile*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln |R_{\text{APG}}(c, n)| = \ln \left(\frac{a}{b} \right)$$

in the asymmetric regime $a > b > 0$, subject to suitable regularity assumptions on the stabilized metric.

The heuristic curvature proxy motivating this conjecture is informed by standard curvature constructions for Riemannian metrics [10, 19, 21]:

$$\tilde{R}_{\text{APG}}(c, n) = -\frac{(a^n - b^n)(a^n + b^n)}{2a^n b^n \ln(a/b)} + O(\lambda).$$

This expression is not presented as a proved scalar curvature formula. A full derivation of the scalar curvature from all non-vanishing Christoffel symbols is left for future work.

6 Coordinate Stability and Extremal Entropy Boundaries

Theorem 2 (Coordinate Stability and Extremal Entropy Boundaries). *Let*

$$w_a = \frac{a^2}{a^2 + b^2}, \quad w_b = \frac{b^2}{a^2 + b^2}.$$

Then

$$H(W) \leq \ln 2,$$

with equality if and only if

$$w_a = w_b = \frac{1}{2},$$

equivalently $a = b$. Moreover,

$$\lim_{a/b \rightarrow \infty} H(W) = 0.$$

Proof. The Shannon entropy of a two-point probability distribution satisfies the standard maximum-entropy bound [28, 8]

$$H(W) = -(w_a \ln w_a + w_b \ln w_b) \leq \ln 2,$$

with equality exactly when the distribution is uniform:

$$w_a = w_b = \frac{1}{2}.$$

Since $a, b > 0$, this equality is equivalent to $a^2 = b^2$, hence $a = b$.

If $a/b \rightarrow \infty$, then

$$w_a \rightarrow 1, \quad w_b \rightarrow 0.$$

Therefore

$$H(W) \rightarrow -(1 \ln 1 + 0 \ln 0) = 0,$$

where $0 \ln 0$ is interpreted as 0 by the standard limiting convention. □

Corollary 1 (Maximum Local Sensitivity). *The first-order deformation coefficient*

$$\frac{H(W)}{2}$$

attains its unique global maximum

$$\frac{\ln 2}{2}$$

if and only if

$$a = b.$$

Proof. By Theorem 2,

$$H(W) \leq \ln 2,$$

with equality if and only if

$$a = b.$$

Dividing by two gives

$$\frac{H(W)}{2} \leq \frac{\ln 2}{2},$$

with equality if and only if

$$a = b.$$

Therefore, the first-order deformation coefficient in Equation (2) attains its unique maximum at the balanced configuration. $\square$

Remark 3. *This theorem gives APG a simple extremal principle: balanced coordinate systems maximize the first-order deformation sensitivity in the local expansion (2), while highly skewed systems minimize it.*

7 The One-Sided Height–Radical Entropy Bound

Following the radical constructions appearing in the *abc*-conjectural literature [20, 22, 30, 15], for a primitive integer triple (a, b, c) , define the radical

$$\text{rad}(abc) = \prod_{p|abc} p.$$

Define the APG Global Information Invariant

$$I_n(a, b, c) = H(W) \ln(\text{rad}(abc)).$$

The absolute logarithmic projective height is the standard Weil projective height used in Diophantine geometry [5, 18, 17, 29]:

$$h(a : b : c) = \ln \max(|a|, |b|, |c|).$$

Theorem 3 (One-Sided Height–Radical Entropy Bound). *Let $(a, b, c) \in (\mathbb{Z}^+)^3$ be primitive and satisfy*

$$a + b = c.$$

Then

$$I_n(a, b, c) \leq 3H(W) h(a : b : c).$$

Proof. Since

$$\text{rad}(abc) \leq abc$$

and $a < c$, $b < c$, we have

$$\text{rad}(abc) \leq abc < c^3.$$

Taking logarithms,

$$\ln(\text{rad}(abc)) \leq 3 \ln c.$$

Since $a + b = c$ with positive coordinates, $h(a : b : c) = \ln c$. Therefore

$$\ln(\text{rad}(abc)) \leq 3h(a : b : c).$$

Multiplying by $H(W) > 0$ gives

$$I_n(a, b, c) = H(W) \ln(\text{rad}(abc)) \leq 3H(W) h(a : b : c).$$

$\square$

Remark 4. *This theorem is intentionally one-sided. A universal lower bound of the form*

$$C_1 h(a : b : c) \leq I_n(a, b, c)$$

is not asserted here.

8 Geometric Dynamics and Moduli-Space Pathways

The geodesic equation associated with the metric g is the standard Riemannian geodesic equation [10, 19]:

$$\frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k \frac{dx^i}{dt} \frac{dx^j}{dt} = 0,$$

where

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}).$$

Proposition 2 (Partial Geodesic Structure). *Let g be the regularized APG metric. A representative Christoffel coefficient is*

$$\Gamma_{cc}^c = \frac{1 - 2n(V_n + \lambda)}{2c^3 \Delta},$$

where

$$\Delta = \left(\frac{n}{c^2} + \lambda \right) (V_n + \lambda) - \frac{1}{c^2}.$$

Proof. The derivation is given in Appendix A. □

Conjecture 2 (APG–Frey/Faltings Height Connection). *Following the Frey–Ribet–Faltings arithmetic interface [14, 24, 13], for the Frey-type curve*

$$E : y^2 = x(x - a^n)(x + b^n),$$

the APG Local Closure Defect may control the stable Faltings height $h_F(E)$ through an inequality of the general form

$$\delta_K(a, b, n) h_F(E) \geq \frac{nH(W)}{24} - O(\ln n),$$

subject to suitable arithmetic and minimal-model hypotheses from elliptic-curve and height theory [29, 27].

Heuristic 1 (Extremal Near-Miss Clustering). *In the broader setting of Diophantine approximation and near solutions [4, 6, 25, 32, 31, 9], for fixed $n > 2$, primitive integer triples producing small closure defect are expected to cluster near the empirical path*

$$\left| \ln \left(\frac{a}{b} \right) \right| \approx \frac{\sqrt{12d_P(c, n)}}{n}.$$

Here, $d_P(c, n)$ denotes the normalized APG defect functional introduced in APG I:

$$d_P(c, n) = \frac{|c^n - a^n - b^n|}{c^n}.$$

9 Computational Validation

To assess the accuracy of the local asymptotic theorem from APG I [1], we compare the exact Local Closure Defect with its first-order entropy approximation for the representative coordinate pair $(a, b) = (3, 4)$. For this pair,

$$c_2 = 5, \quad H(W) = 0.653418.$$

As shown in Figure 1, the first-order APG approximation closely tracks the exact Local Closure Defect near the Euclidean baseline $n = 2$. The divergence observed for larger deformation parameters is consistent with the higher-order remainder term $O(K_P^2)$ in Equation (2).

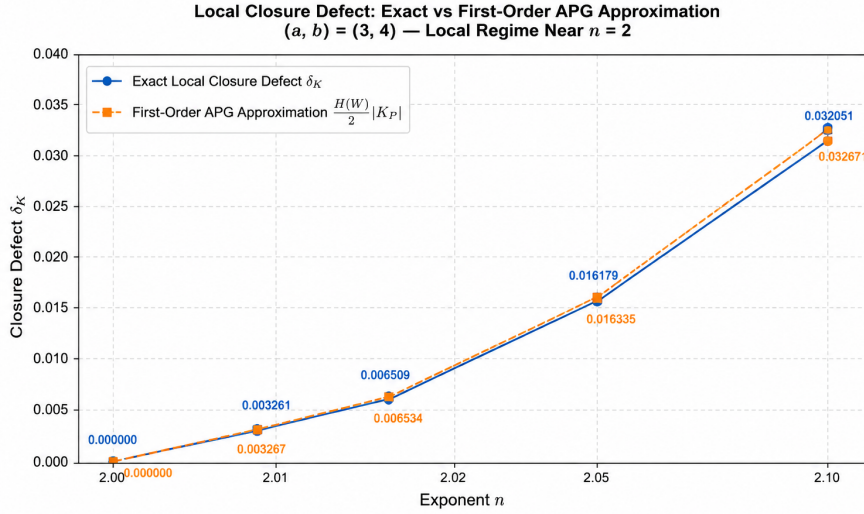


Figure 1: Validation of the local APG asymptotic theorem for the representative coordinate pair $(a, b) = (3, 4)$. The exact Local Closure Defect δ_K is compared with the first-order approximation $\frac{H(W)}{2}|K_P|$. The close agreement near the Euclidean baseline $n = 2$ provides numerical evidence consistent with the local asymptotic expansion, while the increasing deviation for larger values of $|K_P|$ reflects the contribution of higher-order terms beyond the first-order regime.

The numerical values underlying Figure 1 are reported in Table 1. The approximation error remains extremely small near the Euclidean baseline and increases as the exponent moves farther from the local asymptotic regime.

Table 1: Exact defect and first-order APG approximation for $(a, b) = (3, 4)$.

n	K_P	Exact δ_K	$\frac{H(W)}{2} K_P $	Error
2.00	0.00	0.000000	0.000000	0.000000
2.01	0.01	0.003261	0.003267	0.000006
2.02	0.02	0.006509	0.006534	0.000025
2.05	0.05	0.016179	0.016335	0.000156
2.10	0.10	0.032051	0.032671	0.000620
3.00	1.00	0.272000	0.326709	0.054709
4.00	2.00	0.460800	0.653418	0.192618

These computations provide numerical evidence consistent with the asymptotic theorem. They do not constitute an independent analytical proof. The computational script used to reproduce these values is provided in Appendix B.

10 Information-Geometric Parameter Mapping

To illustrate the behavior of the regularized information tensor across the deformation landscape, in the spirit of parameter mappings in information geometry [3, 2], Table 2 reports representative values of the determinant, heuristic curvature proxy, and height-bound coefficient for the canonical coordinate pair $(a, b) = (3, 4)$. The values were generated by direct evaluation of the metric formula in Equation (3). Since the scalar curvature formula remains conjectural, the curvature-related column is reported only as a heuristic proxy.

For the parameter set

$$a = 3, \quad b = 4, \quad c_2 = 5, \quad \lambda = 0.60,$$

the determinant values are computed from

$$\det(g) = \left(\frac{n}{c_2^2} + \lambda \right) (V_n + \lambda) - \frac{1}{c_2^2},$$

where

$$V_n = \frac{a^n b^n (\ln(a/b))^2}{(a^n + b^n)^2}.$$

The heuristic curvature proxy $\tilde{R}_{\text{APG}}$ is evaluated using the ordered asymmetry ratio $4/3$, because the proxy in Conjecture 1 is stated for an ordered asymmetric pair $a > b > 0$.

Table 2: Tensor, heuristic curvature proxy, and height boundary vectors for $(a, b) = (3, 4)$, $c_2 = 5$, $\lambda = 0.60$, and $H(W) = 0.65342$ nats. Since curvature remains conjectural, the curvature column is labeled as a heuristic proxy.

Exponent n	K_P	$\det(g)$	$\tilde{R}_{\text{APG}}$	$C_2 = 3H(W)$
2.00	0.00	0.38097	-2.11219	1.96026
2.05	0.05	0.38215	-2.17092	1.96026
2.10	0.10	0.38334	-2.23009	1.96026
3.00	1.00	0.40443	-3.38654	1.96026
4.00	2.00	0.42748	-4.94311	1.96026

Table 2 shows that the determinant remains positive throughout the sampled deformation range, consistent with Theorem 1. The increasing magnitude of the heuristic curvature proxy qualitatively motivates Conjecture 1, but it is not used as a proved curvature theorem.

11 Conclusion

APG II establishes an information-geometric formulation of exponent deformation spaces. It introduces a regularized tensor framework, proves positive definiteness on stabilized exponent domains, establishes an extremal entropy principle, proves a one-sided height-radical entropy bound, and formulates conjectural directions involving curvature growth and Frey/Faltings heights.

The rigorous contribution of APG II lies in the stabilized metric, the coordinate stability theorem, and the height-radical upper bound. Its arithmetic interfaces remain conjectural and are presented as pathways for future research rather than completed obstruction theorems.

A Derivation of the Partial Christoffel Symbol

Let

$$A = g_{cc} = \frac{n}{c^2} + \lambda, \quad B = g_{cn} = -\frac{1}{c}, \quad C = g_{nn} = V_n + \lambda.$$

Then

$$g = \begin{pmatrix} A & B \\ B & C \end{pmatrix}, \quad \Delta = \det g = AC - B^2.$$

The inverse metric is

$$g^{-1} = \frac{1}{\Delta} \begin{pmatrix} C & -B \\ -B & A \end{pmatrix} = \frac{1}{\Delta} \begin{pmatrix} C & \frac{1}{c} \\ \frac{1}{c} & A \end{pmatrix}.$$

Hence

$$g^{cc} = \frac{C}{\Delta}, \quad g^{cn} = \frac{1}{c\Delta}.$$

Using

$$\Gamma_{cc}^c = \frac{1}{2} g^{cl} (2\partial_c g_{cl} - \partial_l g_{cc}),$$

we compute

$$\partial_c g_{cc} = -\frac{2n}{c^3}, \quad \partial_c g_{cn} = \frac{1}{c^2}, \quad \partial_n g_{cc} = \frac{1}{c^2}.$$

Therefore

$$\Gamma_{cc}^c = \frac{1}{2} [g^{cc} \partial_c g_{cc} + g^{cn} (2\partial_c g_{cn} - \partial_n g_{cc})].$$

Substituting the derivative values gives

$$\Gamma_{cc}^c = \frac{1}{2} \left[\frac{C}{\Delta} \left(-\frac{2n}{c^3} \right) + \frac{1}{c\Delta} \left(\frac{1}{c^2} \right) \right].$$

Thus

$$\Gamma_{cc}^c = \frac{1 - 2nC}{2c^3\Delta} = \frac{1 - 2n(V_n + \lambda)}{2c^3\Delta}.$$

Remark 5. *Earlier simplified formulas for Γ_{cc}^c that ignore the off-diagonal term g_{cn} are not used here. The expression above follows from the full inverse metric.*

B Computational Validation Code

The following Python script reproduces the numerical comparison between the exact Local Closure Defect and the first-order APG approximation reported in Table 1.

```
import numpy as np

def run_large_scale_apg_validation(limit):
    errors = []
    records = []
```

```

for a in range(1, limit):
    for b in range(a, limit):
        if np.gcd(a, b) != 1:
            continue

        a2, b2 = a**2, b**2
        c2_sq = a2 + b2
        wa = a2 / c2_sq
        wb = b2 / c2_sq
        H_W = -(wa * np.log(wa) + wb * np.log(wb))

        K = 0.01
        n = 2.0 + K
        c2 = np.sqrt(c2_sq)

        exact_defect = abs(1.0 - (a**n + b**n) / (c2**n))
        linear_approx = (H_W / 2.0) * K
        abs_error = abs(exact_defect - linear_approx)

        errors.append(abs_error)
        records.append((a, b, exact_defect, linear_approx, abs_error))

return np.mean(errors), np.max(errors), records

mean_err, max_err, records = run_large_scale_apg_validation(100)

print(f"Mean_Absolute_Error: {mean_err:.8f}")
print(f"Max_Absolute_Error: {max_err:.8f}")

```

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